

CSE 423 Computer Graphics

Makeup Quiz (20 marks)

Duration: 20 minutes

Name:	ID:	Section: 10
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1. Can additive color model produce active black color?

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No. Additive models (like RGB) work by emitting light. Black is the pure *absence* of light, so a screen cannot actively emit a "black beam" of light.

2. When we increase the ambient reflection coefficient of an object, the whole object seems to get brighter. But when we increase the diffuse reflection coefficient, one spot gets more brighter than the others. Why?

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Ambient light is directionless and uniform. Increasing its coefficient mathematically adds the exact same amount of brightness to every point on the object.

Diffuse light is directional. Its intensity depends on the angle between the light source | and the surface. Increasing its coefficient acts as a multiplier, meaning surfaces pointing directly at the light get significantly brighter, while surfaces angled away remain darker.

3. Given inclination angle β and skewness angle α , prove the oblique projection formulae for the projection plane $XY / Z = 0$ 10

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